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Phase-space Analysis of Ordered and Disordered Nonthermal Ion Energization during Magnetic Reconnection

Young Dae Yoon[1,2], Paul M. Bellan[3], and Gunsu S. Yun[2,4]
[1] Asia Pacific Center for Theoretical Physics, Pohang, Gyeongbuk 37673, Republic of Korea; youngdae.yoon@apctp.org
[2] Department of Physics, POSTECH, Gyeongbuk 37673, Republic of Korea
[3] Department of Applied Physics, California Institute of Technology, Pasadena, CA 91125, USA
[4] Division of Advanced Nuclear Engineering, POSTECH, Gyeongbuk 37673, Republic of Korea
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Abstract

Anomalous ion heating is frequently observed to accompany magnetic reconnection, yet there is little consensus on its origin. Instead of the usual velocity-space analysis, we use phase-space analysis to exhaustively explain how ions are nonthermally energized during collisionless, antiparallel magnetic reconnection. There are both ordered and disordered aspects in the process; the former is explained in terms of conservative quantities, and the latter is explained by demonstrating chaos through a direct calculation of Lyapunov exponents. The former induces "multibeam-like heating" in all three directions, whereas the latter induces stochastic bulk heating. Profiles of the ion temperature tensor components during reconnection can be easily understood by the phase-space distributions of ions in different motional stages.

Unified Astronomy Thesaurus concepts: Solar magnetic reconnection (1504); Solar coronal heating (1989)

1. Introduction

Magnetic reconnection is a crucial plasma process in which a change in magnetic field topology converts the field energy into particle kinetic energy. Ion energization is an important element of such a conversion, but the nature of ion heating during reconnection typically cannot be explained by classical heating mechanisms, i.e., collisional Ohmic or viscous heating (Hsu et al. 2000; Hare et al. 2017). The heating is therefore anomalous, and this anomaly seems to be a property intrinsic to collisionless reconnection, judging from the ubiquity of such ion heating observations in a variety of plasma contexts such as in solar, magnetospheric, and laboratory environments (Ono et al. 1996, 2011; Priest et al. 1998; Hsu et al. 2000; Lin et al. 2003; Emslie et al. 2004; Stark et al. 2005; Fiksel et al. 2009; Magee et al. 2011; Yoo et al. 2013; Chai et al. 2016; Hare et al. 2017).

Previous studies have suggested various origins for such heating, including pickup (Drake et al. 2009a, 2009b; Knizhnik et al. 2011; Drake & Swisdak 2014), turbulence (Matthaeus et al. 1984), electric field fluctuations (Fiksel et al. 2009), remagnetization (Yoo et al. 2013), shocks or viscous damping (Ono et al. 2011), and stochastic heating (Yoon & Bellan 2018, 2019, 2020). Other studies have conducted ad hoc investigations of ion motion (Hoshino et al. 1998; Aunai et al. 2011; Zenitani et al. 2013). However, despite the extensive amount of relevant research, an unambiguous consensus on the ion heating mechanism has yet to be settled.

Most previous studies have mainly focused on local velocity-space analyses at multiple spatial locations (Hoshino et al. 1998; Shay et al. 1998; Drake et al. 2009b; Aunai et al. 2011; Zenitani et al. 2013; Haggerty et al. 2015; Usami et al. 2017), which facilitate comparisons to observational data. However, collisionless reconnection involves ion kinetic scales where individual ion oscillatory motion becomes important, and so the distribution function is translationally variant. In such cases, examination of phase-space distributions, which simultaneously reveal both single-particle orbital dynamics and collective dynamics, can provide a more advantageous approach. This advantage was demonstrated in recent investigations of current sheet equilibration processes (Yoon et al. 2021, 2023).

As an illustrative simple example, consider a distribution of particles under a simple harmonic potential $V(x) = x^2/2$ with an average energy H_0 and energy spread ΔH (Figure 1(a)). The Hamiltonian of the system is $H(x, v) = (x^2 + v^2)/2$, so the distribution $f = f(H) \sim \exp\left(-\left(\sqrt{x^2 + v^2} - \sqrt{2H_0}\right)\Big/\Delta H\right)$ is a stationary solution (Figure 1(b)). A common approach is to examine velocity-space distributions at selected spatial locations (Figure 1(c)) and the spatial profile of the temperature $T \sim \int v^2 f dv$ (Figure 1(d)). The example is in 1D, but in 2D or 3D, parallel and perpendicular velocity distributions and temperatures are often examined. However, it is difficult to correctly assess the physical situation solely by examining Figures 1(c) and (d); only by examining the phase-space distribution in Figure 1(b) can one easily realize that the system is a thermal distribution of particles circulating in phase-space in a simple harmonic potential.

In this paper, we use phase-space analysis to elucidate how ions are nonthermally energized in all three directions during 2D antiparallel collisionless magnetic reconnection. Leveraging the symmetry of the system, we describe ion phase-space distributions in terms of conserved quantities and effective potentials. We then perform Monte Carlo test-particle simulations to examine different stages of ion motion and compare them to the phase-space distributions, as well as to directly calculate Lyapunov exponents that quantify the existence of chaotic motion. There are both ordered and disordered aspects of ion motion, with the former being responsible for "multibeam-like" heating in all three directions and the latter being responsible for the rapid, stochastic ion heating. Ion temperature profiles during reconnection can be well explained by the

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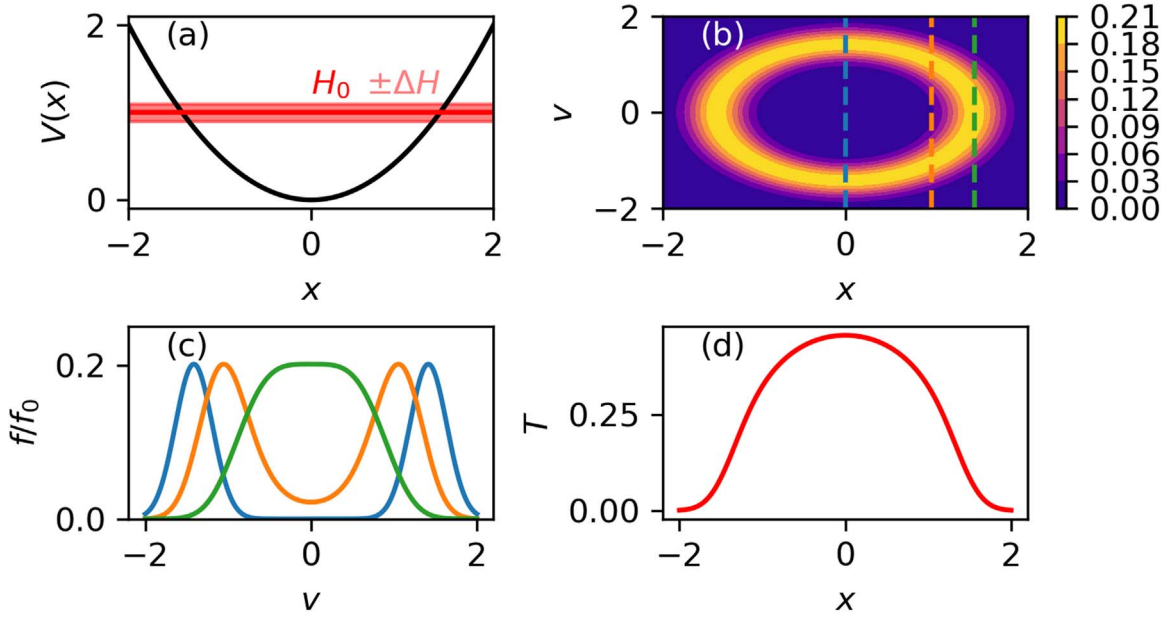


Figure 1. (a) A simple harmonic potential (black). Also shown are the average energy $H_0 = 1$ (red line) and energy spread $\Delta H = 0.1$ (red shaded area) of a hypothetical distribution of particles. (b) An equilibrium distribution in (x, v) space. (c) Cuts across the dashed lines in (b). (d) Spatial profile of $T = \int v^2 f dv$.

phase-space distributions and their classification into different ion motional stages.

2. Setup

In order to analyze ion phase-space dynamics, 2D/3V particle-in-cell (PIC) simulations were conducted using the open-source SMILEI code (Derouillat et al. 2018). The initial setup was the Harris equilibrium (Harris 1962; Birn et al. 2001), i.e., with magnetic field $\boldsymbol{B} = \hat{y}B_0 \tanh(x/\lambda)$, ion/electron density profiles $n_i = n_e = n_0 \cosh^{-2}(x/\lambda)$, and constant ion/electron temperatures $T_i = T_e = T$ that satisfy $n_0(T_i + T_e) = B_0^2/2\mu_0$. The domain was $(x, y) = (1024, 8192) = (51.2, 409.6)d_i$ where $d_i = c/\omega_{pi}$ is the ion skin depth. The mass ratio was $m_i/m_e = 100$, the Alfvén velocity was $v_A/c = 0.025$, $\lambda = d_i$, and particles per cell ranged from 200 to 1000 (density dependent). Boundary conditions were periodic in the y-direction and open in the x-direction. Reconnection was initiated by a Gaussian perturbation.

Figure 2 shows the electric field normalized by $v_A B_0$ and ion temperature tensor components normalized by $m_i v_A^2$ at $t_{PIC} = 5600\omega_{pi}^{-1}$. Black lines are magnetic field lines. Two plasmoids have formed at $y < 0$, so we will focus on the $y > 0$ region. As seen previously (Drake et al. 2009b; Aunai et al. 2011; Yoo et al. 2013), the in-plane Hall electric fields point toward the $x = 0$ axis (Figure 2(a)) and away from the separatrix in the y-directions (Figure 2(b)). The reconnection electric field is constant throughout the reconnecting region (Figure 2(c)). The temperature tensor components (Figures 2(d)–(f)) were calculated as $T_{\sigma\sigma} = (m_i/n_i)\int v'_\sigma v'_\sigma f_i dv'_\sigma$ where v'_σ is the random part of the velocity in the σ-direction and $f_i = f_i(\boldsymbol{x}, \boldsymbol{v}, t)$ is the ion distribution function. Quadrupole Hall magnetic fields are also present (Figure 2(g)).

The Hamiltonian of an ion in 2D ($\partial/\partial z \to 0$) is, after normalizing length to λ, time to $\omega_c^{-1} = (q_i B_0/m_i)^{-1}$, mass to m_i, and magnetic field to B_0,

$$H = \frac{1}{2}(v_x^2 + v_y^2) + \frac{(p_z - A_z)^2}{2} + \phi, \tag{1}$$

where $A_z(x, y, t)$ and $\phi(x, y, t)$ are vector and electrostatic potentials and $p_z = v_z + A_z$ is a constant. Near the origin, the in-plane magnetic fields during reconnection have the dependences $B_y \sim x$ and $B_x \sim b_r y$, where $b_r \ll 1$ is the relative magnitude of the reconnected field. There is also a constant E_z throughout the reconnection region (Figure 2(c)), so

$$A_z \simeq \frac{1}{2}(-x^2 + b_r y^2) - E_z t. \tag{2}$$

Now, assuming the in-plane electric fields are largely due to the Hall effect, $(\boldsymbol{E} \cdot \boldsymbol{B})_{\text{in-plane}} = 0$, so $\boldsymbol{B}$ field lines are equipotentials, i.e., $\phi(x, y) \sim E_H(x^2 - b_r y^2)/2$ where E_H is the Hall electric field's reference magnitude. The Hamiltonian is now $H = \frac{1}{2}(v_x^2 + v_y^2) + \chi(x, y, t)$, where

$$\chi(x, y, t) \simeq \frac{(p_z + E_z t + (x^2 - b_r y^2)/2)^2}{2} + \frac{E_H}{2}(x^2 - b_r y^2) \tag{3}$$

is the 2D effective potential. Most ions have negative p_z because $p_z = v_z - (x^2 - b_r y^2)/2 < 0$ at upstream (small y and large x).

3. Phase-space Analysis

3.1. Inflow Direction

Let us first examine ion dynamics in (v_x, x) space at $t_{PIC} = 5600\omega_{pi}^{-1}$. At a given y and t, Equation (3) shows that $\chi(x) \sim x^4 + 2(2p_z + E_H + 2E_z t - b_r y^2)x^2 + g(y, t)$. If the coefficient of x^2 is positive, then $\chi(x)$ is single welled (V shape), but if it is negative, then $\chi(x)$ is double welled (W shape). For a given t, increasing y^2 always results in a double-

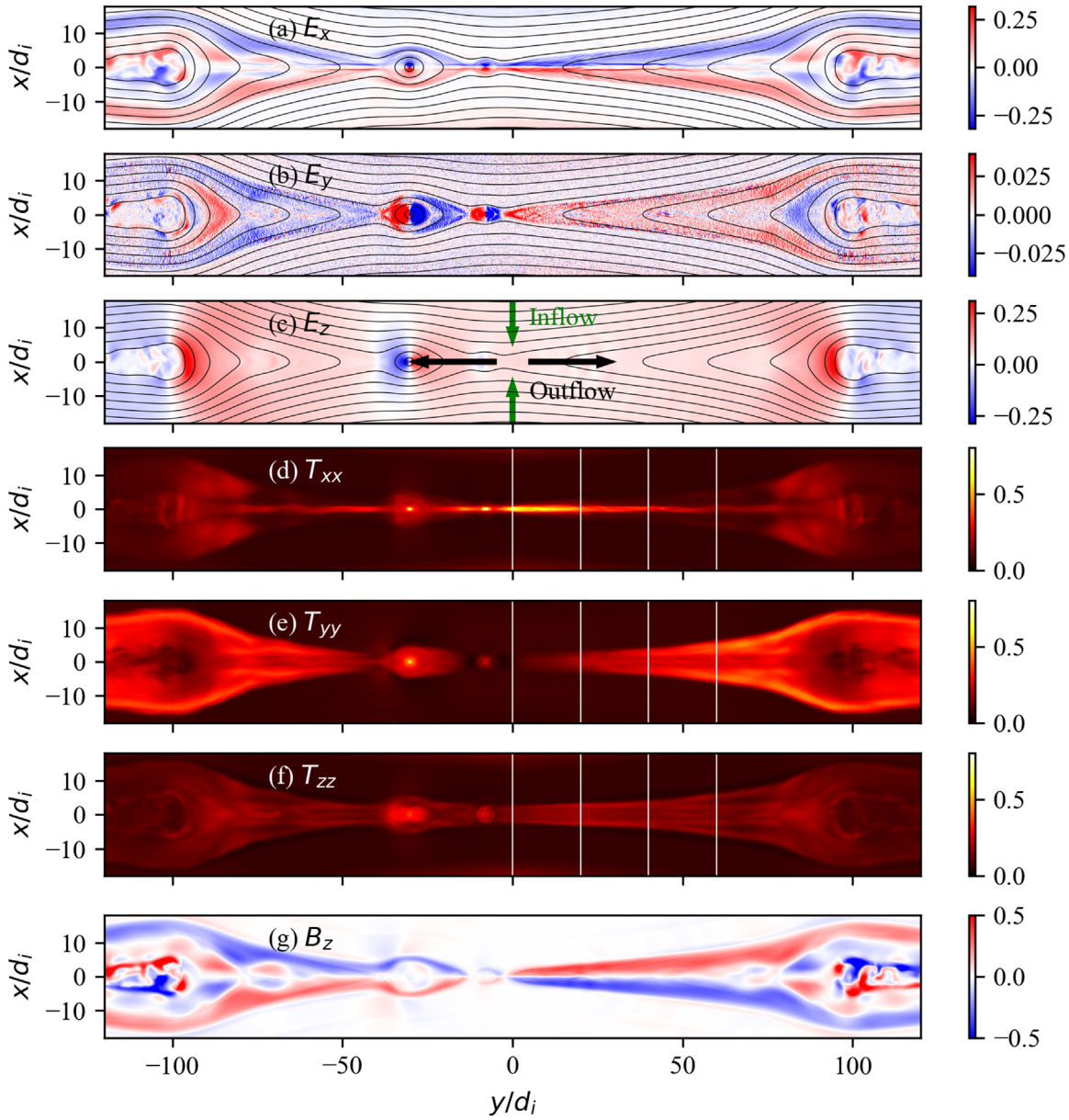


Figure 2. Profiles of the electric field, ion temperature tensor components, and the out-of-plane magnetic field from the PIC simulation.

well $\chi(x)$. Also, finite v_y is just an offset to H and can be set to zero when examining (v_x, x) space.

The colors in Figure 3 show that the log-scale ion distribution function in (v_x, x) space at $y = 0$, 20, 40, and $60d_i$ (white lines in Figures 2(d)–(f)) exhibit phase-space holes. The colored solid lines in Figure 3 show representative ion phase-space contours, i.e., $v_x = \pm\sqrt{2(H - \chi)}$, calculated from Equation (1) with local ϕ and A_z profiles from the simulation for varying values of p_z and H, and the dotted lines are the corresponding $\chi(x)$ with the $v_x = 0$ line acting as H. The morphology of the phase-space hole is almost completely explained by dynamics in the x-direction and so by the shape of $\chi(x)$.

For small y (Figures 3(a)–(c)), $\chi(x)$ is single welled due to the strong E_H. The corresponding phase-space contour is circular about the origin. Further downstream (Figure 3(d)), y becomes big enough so that $\chi(x)$ is double welled for some ions (cyan dotted lines). In this case the ions are trapped about a finite x (cyan solid lines).

The inset in Figure 3(a) shows the log-scale v_x distribution at $x = 0$ (blue) and $x = 4$ (orange), respectively corresponding to the dashed cuts. Large T_{xx} in Figure 2(d) is "multibeam-like" because the high temperature comes from phase-space circulation with a huge void around $(v_x, x) = (0, 0)$, which at $x = 0$ are two counterpropagating beams (Drake et al. 2009b; Aunai et al. 2011). T_{xx} of the individual beams is in fact lower than the upstream T_{xx}. Throughout this paper, the term "multibeam-like" will be used to describe high temperatures that are due to multiple, distinguishable populations rather than a single bulk population.

3.2. Out-of-plane Direction

Next, let us examine ion dynamics in (v_z, x) space. By definition of p_z,

$$v_z(x, y, t) = p_z - A_z \simeq p_z + E_z t + \frac{1}{2}(x^2 - b_r y^2), \qquad (4)$$

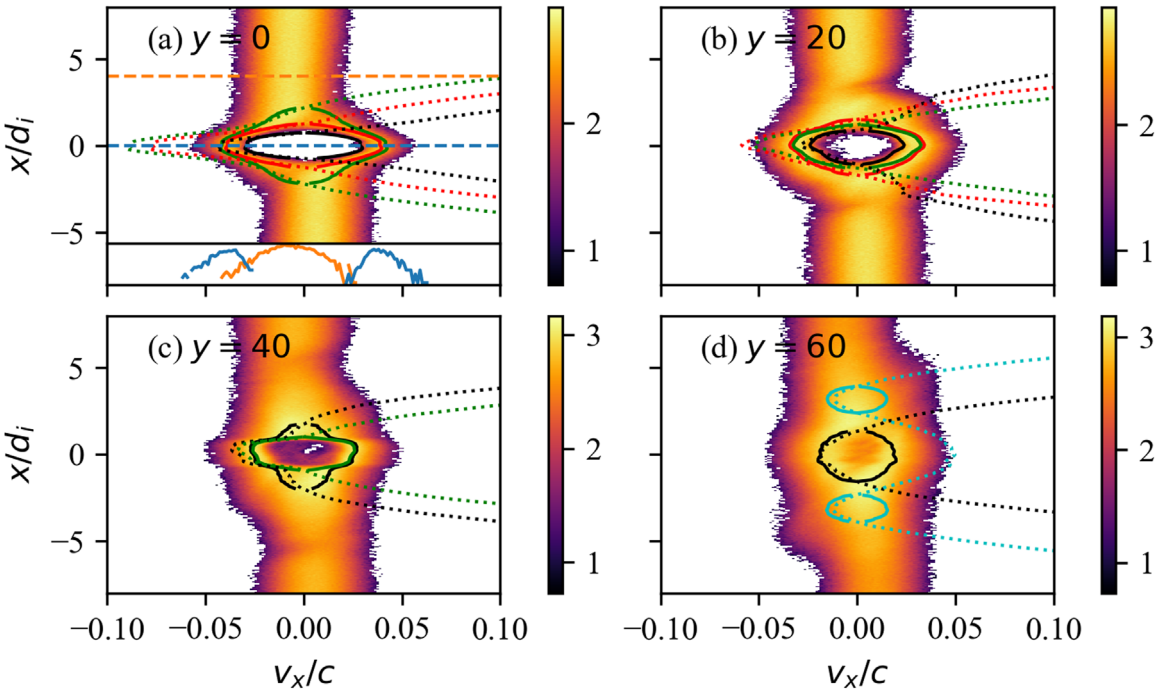


Figure 3. Log-scale ion distributions in (v_x, x) space at different y at $t_{\rm PIC}\omega_{pi} = 5600$. The dotted lines in each panel are $\chi(x)$ for varying values of p_z and H, and the solid lines are corresponding phase-space trajectories. The $v_x = 0$ line acts as H for each χ. The inset in (a) shows cuts of the distribution function across the dashed lines.

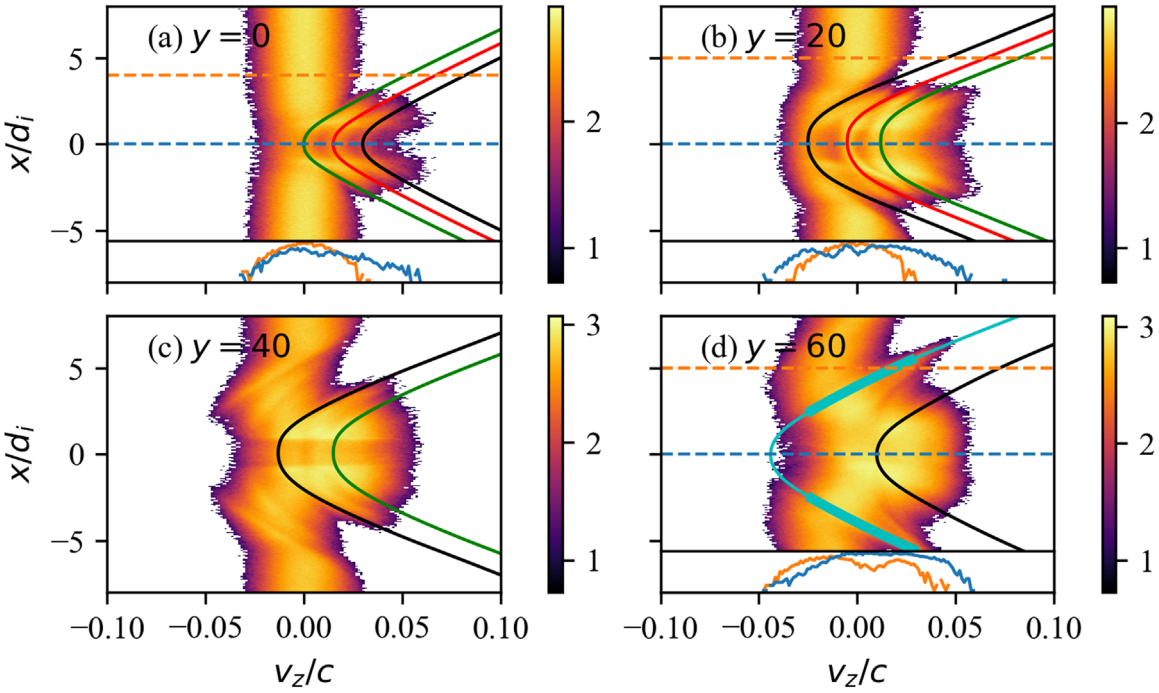


Figure 4. Same as Figure 3, but in (v_z, x) space. The colored lines now represent $v_z(x)$ for different p_z in Equation (4). The insets in (a), (b), and (d) show cuts of the distribution function across the dashed lines.

so $v_z \sim x^2 + i(y, t)$. Equation (4) shows that $i(y, t)$, and so v_z increases linearly in t but decreases quadratically in y.

Figure 4 is the same as Figure 3, except in (v_z, x) space. For $y \leqslant 20d_i$ (Figures 4(a), (b)), $b_r y^2$ is small enough so that $i(y, t) \simeq i(t) \sim t$. The colored solid lines in each panel show $v_z(x) = p_z - A_z(x)$ with different values of p_z. The agreement between the striations and the solid lines in Figures 4(a), (b) confirms the predicted relation $v_z \sim x^2 + i(t)$; the ions are being constantly accelerated by E_z while oscillating in the x-direction.

Further downstream ($y \geqslant 40d_i$; Figures 4(c), (d)), the ions have large enough y that $b_r y^2$ becomes significant. So, Equation (4) shows that the ions decelerate in the z-direction. The ions decelerate back to around $v_z = 0$ and are trapped in double-welled $\chi(x)$ (e.g., populations that oscillate about finite x along the thick cyan line in Figure 4(d)).

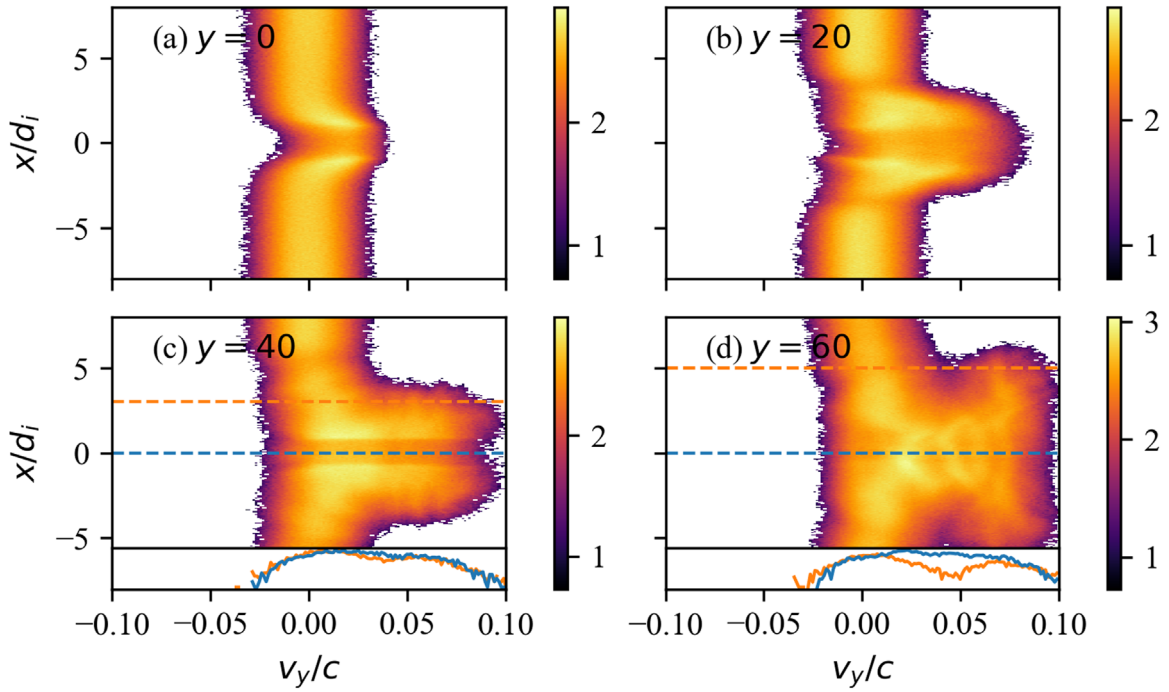


Figure 5. Same as Figure 3 but in (v_y, x) space.

The structure of T_{zz} (Figure 2(f)) is explained by the (v_z, x) space distributions. In Figure 4(d), there are two peaks in the velocity distribution at $x = 5d_i$ (orange line), one due to the inflowing ions and the other due to the double-welled ions. This leads to a multibeam-like increase in T_{zz} near the separatrix. Near $x = 0$, the increase in T_{zz} is also due to multiple beams (blue line in Figure 4(b) inset) that are generated by continuous acceleration by E_z although the beams are phase mixed further downstream (blue line in Figure 4(d) inset).

3.3. Outflow Direction

Next, let us examine ion dynamics in (v_y, x) space (Figure 5). The ions are collectively accelerated as they travel downstream; the source of the acceleration is different at different stages, as will be seen later. The ions are also rapidly phase mixed so barely any striations are visible. As will be seen later, this phase mixing is mainly due to disordered, stochastic motion, as predicted in Yoon & Bellan (2018, 2019). Stochastic heating occurs when a strong perpendicular electric field renders particle motions chaotic, effectively heating up the particles (Karney & Bers 1977; McChesney et al. 1987; Sanders et al. 1998; Stasiewicz et al. 2000; Bourdier et al. 2005; Chandran et al. 2010, 2013; Vranjes & Poedts 2010; Bourouaine & Chandran 2013). The condition is

$$\frac{m_i}{q_i |\boldsymbol{B}|^2} |\nabla_\perp \cdot \boldsymbol{E}_\perp| \equiv \alpha > 1, \tag{5}$$

where $\perp$ means perpendicular to the local magnetic field $\boldsymbol{B}$. One way to understand α is that it is the ratio of the strength of the polarization drift $|v_p|$ to the $\boldsymbol{E} \times \boldsymbol{B}$ drift $|v_E|$, i.e., $\alpha \sim |v_p|/|v_E|$. Therefore, $\alpha > 1$ means that particle drift theory breaks down and should not be used and that magnetic moment μ is not conserved. This condition is intrinsically satisfied in antiparallel reconnection, and α can be as big as m_i/m_e (Yoon & Bellan 2018).

The inset in Figure 5(d) shows that there are two distinct populations (orange line) at $(x, y) = (5, 60)d_i$. The slower population corresponds to inflowing ions at that point in space. The faster population corresponds to ions that have accelerated from upstream and then veered off to large x due to the double-welled $\chi(x)$. The apparent sharp increase of T_{yy} near the separatrix (Figure 2(e)) is multibeam-like because it is due to two completely disparate ion populations.

4. Monte Carlo Analysis

To further understand ion phase-space dynamics and demonstrate chaotic behavior, Monte Carlo test-particle simulations were conducted. The colored lines in Figure 6(a) are trajectories of 200 test ions whose initial velocities were sampled randomly from a 3D Maxwellian distribution with a thermal velocity of $0.01v_A$ and a shift of $-0.006v_A$ in the v_x direction. The ions were placed within $\pm 0.1d_i$ of $(x, y) = (7, 15)d_i$ to delineate disordered motion. Their motion was integrated for $t\omega_{pi} = 2500$ using the Boris algorithm (Birdsall & Langdon 1985) under the electromagnetic fields from the PIC simulation at $t_{\rm PIC}\omega_{pi} = 5600$. The trajectories of the ions were classified into different ion motional stages based on their colors (green, orange, blue, and red), representing different time intervals. The overlaid gray colors show a representative $\chi(x, y)$ calculated from the PIC simulation at $t_{\rm PIC}\omega_{pi} = 5600$ for $p_z + E_z t = 10$, which confirms the shape in Equation (3). The single-well to double-well transition of $\chi(x)$ for increasing y is clear. Figure 6(b) shows $\log_{10}\alpha$ (color) and $\boldsymbol{B}$ lines from the PIC simulation, and the stochastic condition is satisfied in regions where $\alpha > 1$. The cyan contour corresponds to $\alpha = 1$. α surpasses unity across the separatrix and reaches $100 = m_i/m_e$ near $x = 0$. The ions' trajectories in (v_x, x), (v_y, x), and (v_z, x) spaces are shown in Figures 6(c), (d), and (e), respectively.

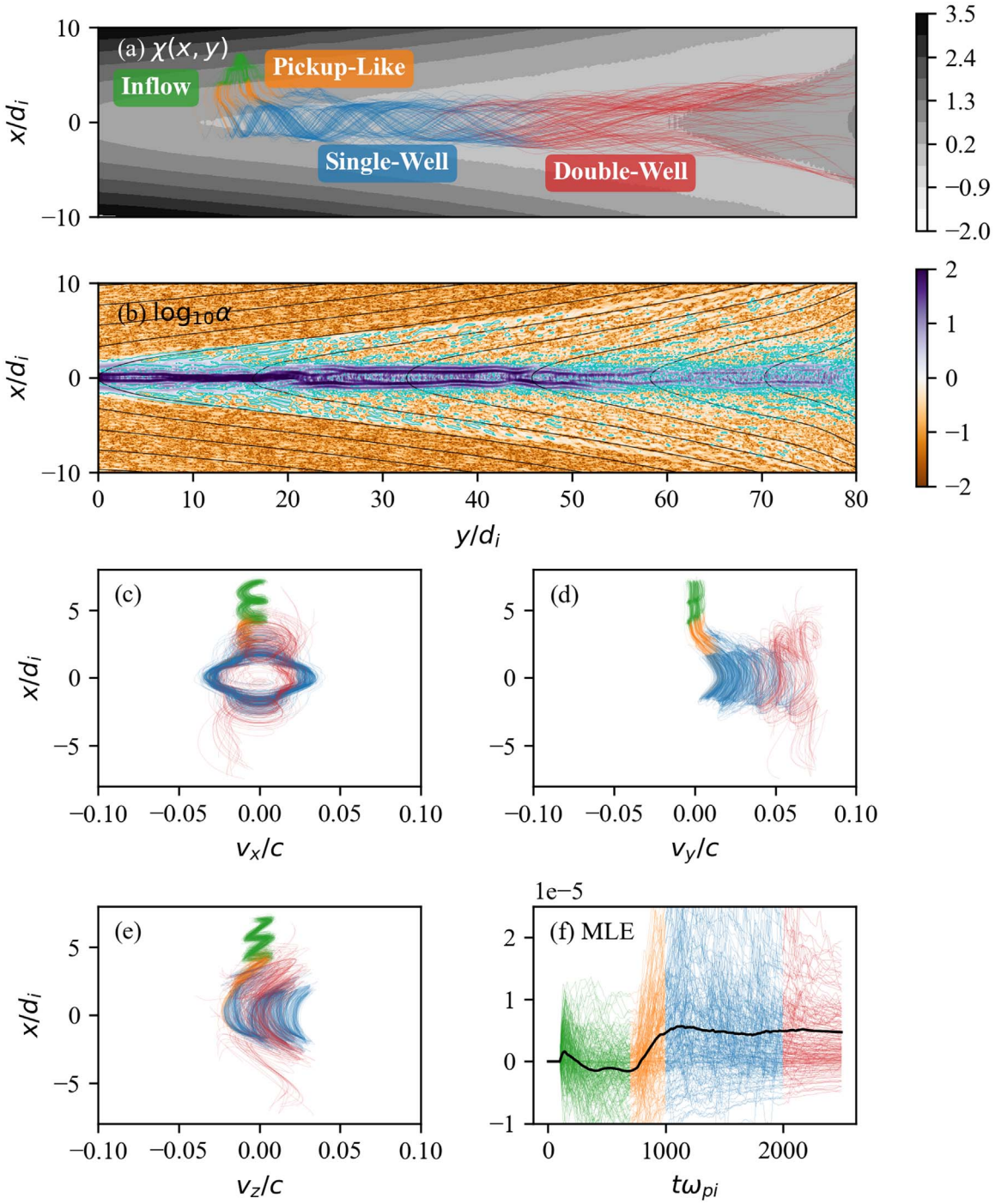


Figure 6. (a) A representative $\chi(x, y)$ (gray colors) and ion trajectories from the Monte Carlo test-particle simulation. (b) Magnetic field lines overlaid on $\log_{10}\alpha$ (color). The cyan contour is $\log_{10}\alpha = 0$. (c), (d), (e) Ion phase-space trajectories in (v_x, x), (v_y, x), and (v_z, x) spaces, respectively. (f) MLEs of all ions and their average value (black line). The green, orange, blue, and red line colors correspond to the inflow, pickup-like, single-well, and double-well stages, respectively.

The ions' maximal Lyapunov exponents (MLEs) were simultaneously calculated using the Benettin algorithm (Benettin et al. 1980). MLE measures the exponential separation of infinitesimally close trajectories and is indicative of chaotic behavior when positive. Figure 6(f) shows the MLEs and their mean value (black line). MLEs at early times ($t\omega_{pi} = 0$–100) were discarded as is customary for the algorithm.

Ion dynamics at each motional stage will now be dissected. At the inflow (green) stage, ions collectively $(E_z\hat{z}) \times (B_y\hat{y})$ drift toward the separatrix. The mean MLE converges to a nonpositive value (Figure 6(f)) indicating non-chaotic behavior, as is expected because $\alpha < 1$ in this region.

At the pickup-like (orange) stage, the ions cross the separatrix and start to $(E_x\hat{x}) \times (B_z\hat{z})$ drift in the outflow direction (acceleration by Hall E_x and rotation by quadrupole B_z). This is the pickup-like mechanism (Drake et al. 2009b), but this motion pertains to less than a full gyration and only provides the initial outflow acceleration (Figure 6(d)). As the ions marginally satisfy $\alpha > 1$, the mean MLE starts to increase (Figure 6(f)).

At the single-well (blue) stage, the ions are trapped in single-well $\chi's$ (Figure 6(c)), which are equivalent to meandering orbits (Speiser 1965). The increasing v_z due to E_z (Figure 6(e)) couples with B_x and, together with E_y, accelerates ions in the outflow direction (see Appendix for details on which acceleration terms dominate ion dynamics in each direction at each motional stage). They also encounter regions of very high α (Figure 6(b)), and their MLE is positive (Figure 6(f)). The ions are thus phase-space mixed and stochastically heated (Figures 6(d), (e)).

At the double-well (red) stage, the ions are trapped at finite x in their now-double-welled $\chi(x)$'s (Figure 6(c)). Their mean v_z returns back to 0 and their outflow acceleration stalls. Therefore, E_z does no net work on the ions in the z-direction and only contributes to the outflow via coupling with B_x. Only stochastic heating effectively broadens the v_z distribution.

Because the ions cross the separatrix at different locations, all four motional stages coexist far downstream. Therefore, the phase-space trajectories in Figures 6(c)–(e) conform well to the distributions in Figures 3(d), 5(d), and 4(d). The temperature profiles in Figures 2(d)–(f) can also be understood in terms of the proportion of ions in each motional stage.

The ergodicity of the ion motion is another observation in support of chaos. In Figures 6(c)–(e), one can see that the test ions reach nearly all points in the phase-space distribution even though they were initially placed in a compact region of phase space.

Note that a real-time test-particle simulation using the electromagnetic (EM) fields at PIC calculation time was also conducted, but only the test-particle simulation using EM fields at a fixed time is shown here because of the difficulty in calculating and calibrating MLE in real time. Nonetheless, the two simulations show similar qualitative behavior; the morphological similarity between the test-particle phase-space trajectories and the phase-space distributions from the PIC simulation is a testament to this fact. Because the Hall electric field grows with time, the real-time particles are accelerated and phase mixed more rapidly and significantly than the test particles.

5. Discussion

Although the exact partition of the available magnetic energy to ion energy requires further investigation, the expected temperature scaling due to stochastic heating can be derived. Energy gain from stochastic heating is proportional to the electrostatic potential ϕ, and the in-plane Hall electric field has the approximate relation $E_x \sim -\partial B_y^2/\partial x$ or $\phi \sim B_y^2$ (Li & Horiuchi 2008; Yoo et al. 2013; Yoon & Bellan 2018), so the ion temperature gain $\Delta T_i \propto B_y^2$. This is consistent with the observed ΔT_i statistics of reconnection events in space (Drake et al. 2009b; Phan et al. 2014).

Another important topic for future investigation is the relative partition of ordered energization and stochastic heating and its temporal/spatial variation. This would be done by examining the phase-space distributions and identifying multiple peaks due to multibeam-like heating and broadening due to stochastic heating. The relative contribution of each distribution structure would then be quantified. For instance, the distribution in Figure 3(a) is a bulk-energetic population circulating in phase space with a lower temperature than the upstream temperature. As we go further downstream, the temperature increases while the v_x amplitude decreases (Figure 3(b)). A means to quantify this distinction should be developed and applied in future analyses.

Drake et al. (2009b) proposed that the pickup mechanism may be grossly responsible for ion acceleration even during antiparallel reconnection. There are several problems with this explanation in the zero- to low-guide-field case. First, the electric field in pickup is due to fast-flowing ions, i.e., $\boldsymbol{E} = -\boldsymbol{v}_i \times \boldsymbol{B}$ (Möbius et al. 1985; Luhmann 2003), which requires a preestablished accelerated flow in the first place, hinting to circular reasoning if applied to reconnection where there is no initial ion flow. In contradistinction, the relevant electric field in reconnection is mainly provided by Hall and pressure effects, which are controlled mostly by electron physics.

Second, pickup predicts an initial ring distribution, which is not observed in our simulations. Diffusion mechanisms invoked in heliospheric pickup to flatten the ring distribution, such as magnetic and adiabatic coolings (Fahr 2007), are too slow to explain the fast phase-space mixing in reconnection.

Third, the nonconservation of the ion magnetic moment in pickup is because neutral particles do not have any, while the newborn ions are born with one. Afterwards, the ions retain their magnetic moments; in fact, their retention is the basis of magnetic cooling (Fahr 2007). In reconnection, however, the nonconservation is due to higher-order drifts becoming more significant than $\boldsymbol{E} \times \boldsymbol{B}$ drifts, thus breaking down guiding-center hierarchy and triggering stochastic heating.

Fourth, even in the strong-guide-field case where pickup is expected to play a role (Drake et al. 2009b; Drake & Swisdak 2014; Usami et al. 2017, 2019), the scaling of the temperature with the guide-field strength is found to be inconsistent with simulations (Usami et al. 2017, Section IIIC). Pickup predicts an increase in temperature with the guide field, whereas stochastic heating correctly predicts a decrease (Yoon & Bellan 2019).

One plausible explanation for the guide-field scaling is that extreme ion heating is due to stochastic heating that disappears at high guide fields, at which point the pickup-like mechanism becomes responsible for more moderate heating (Drake et al. 2009b).

6. Conclusion

In conclusion, ordered and disordered aspects of nonthermal ion energization were revealed through phase-space analysis and conservative quantities. Ion motion can be classified into different stages, and the ion phase-space distribution can be represented by the relative proportion of ions at each motional stage. Ordered energization by in-plane Hall electric fields leads to multibeam-like heating of the ions; the reconnection electric field does zero net work. Disordered energization is achieved through stochastic heating, verified through a direct demonstration of chaos by positive Lyapunov exponents. Ion temperature tensor components can be easily explained by the ion phase-space distributions.

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Appendix
Acceleration Mechanisms at Each Motional Stage

Figure 7(a) shows the mean v_x of the 200 Monte Carlo test ions (black line) and the mean values of different components (brown, purple, and green lines) that contribute to v_x, i.e.,

$$v_x - v_0 = \int E_x dt + \int v_y B_z dt - \int v_z B_y dt, \tag{A1}$$

under appropriate normalization. The local $\boldsymbol{E} \times \boldsymbol{B}$ drift velocity is also shown (pink line). The background colors (green, orange, blue, and red) represent different ion motional stages (inflow, pickup-like, single-well, and double-well, respectively). Figures 7(b), (c) are corresponding plots in the y- and z-directions. Recall that x is the inflow direction, y is the outflow direction, and z is the out-of-plane direction.

At the inflow (green) stage, the ion motion can be described as an $\boldsymbol{E} \times \boldsymbol{B}$ drift in the $-x$-direction, as can be seen by the negative $(\boldsymbol{E} \times \boldsymbol{B})_x$ in Figure 7(a) and the cancellation between $\int E_z dt$ and $\int v_x B_y dt$ in Figure 7(c) (and so $v_x = -E_z/B_y$).

At the pickup-like (orange) stage, the ion motion can still be described as an $\boldsymbol{E} \times \boldsymbol{B}$ drift. In Figure 7(b), v_y, $(\boldsymbol{E} \times \boldsymbol{B})_y$, and $-\int v_x B_z dt$ all overlap, so the outflow acceleration is entirely due to acceleration by E_x and rotation by B_z. $\int E_y dt$ is canceled by $\int v_z B_x dt$. Acceleration by E_x also couples with B_y and contributes to negative v_z acceleration, as can be seen in Figure 7(c). This corresponds to $v_z \sim p_z + x^2/2$ (Equation (4) in the main text), and so if x becomes small, $v_z \sim p_z < 0$. The coupling of E_x with both B_y and B_z can also be seen in Figure 7(a), where $\int E_x dt$ is canceled by the sum of $\int v_y B_z dt$ and $-\int v_z B_y dt$.

At the single-well (blue) stage, ion motion is completely incompatible with $\boldsymbol{E} \times \boldsymbol{B}$ motion, as can be seen by the large discrepancies between the black and pink lines in Figures 7(b) and (c). From this point onwards, there is no average acceleration in the x-direction (Figure 7(a)). In the outflow direction (Figure 7(b)), $-\int v_x B_z dt$ no longer contributes to acceleration in contrast to the pickup-like stage. Instead, the outflow acceleration is due to a combination of $\int E_y dt$ (direct acceleration by the Hall field E_y) and $\int v_z B_x dt$ (acceleration by the reconnection field E_z and rotation by B_x). In the out-of-plane direction (Figure 7(c)), $\int v_x B_y dt$ stops contributing to v_z acceleration, and only $\int E_z dt$ contributes. Also, as the ions are

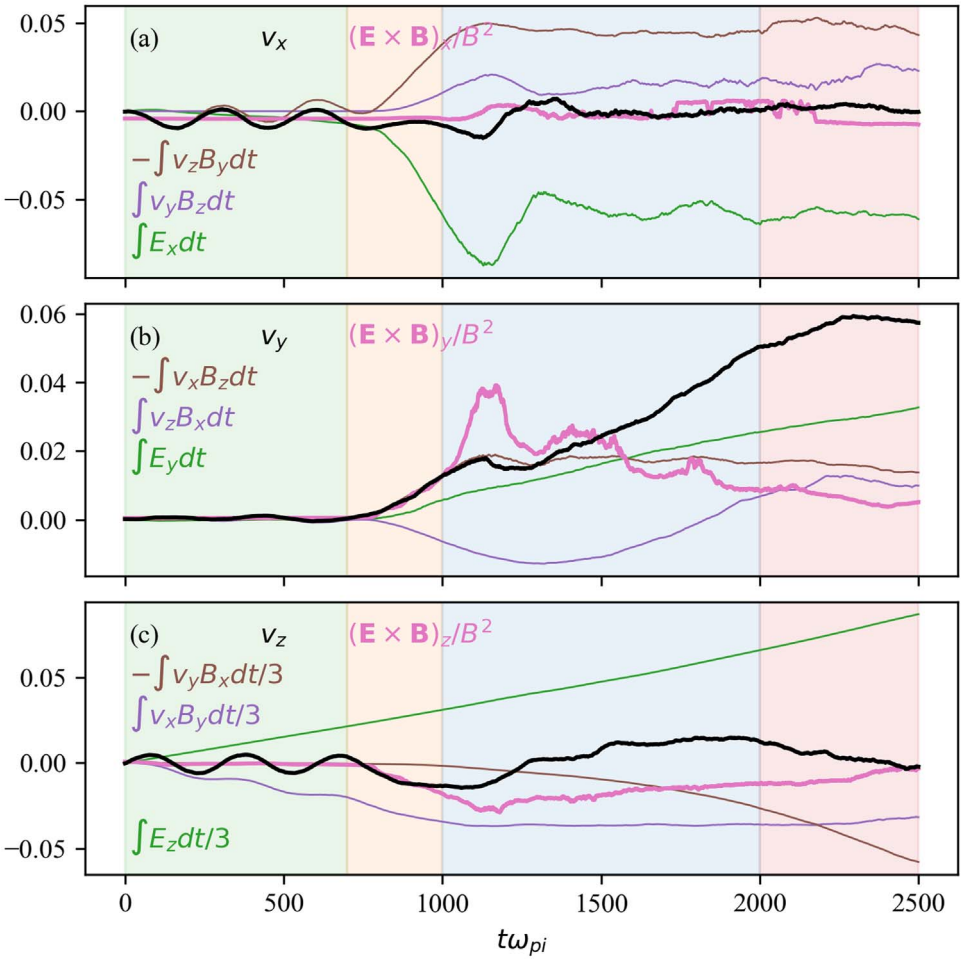


Figure 7. (a) Mean v_x (black line) of the 200 Monte Carlo test ions and the mean values of different components (colored lines) that contribute to v_x. (b) Same for v_y. (c) Same for v_z.

accelerated in the outflow, the increased v_y starts to couple with B_x and contributes to v_z deceleration (brown line).

At the double-well (red) stage, $-\int v_y B_x dt$ overcomes acceleration by E_z and decelerates the ions back to $v_z = 0$ (Figure 7(c)). This decreased v_z in turn decreases $\int v_z B_x dt$, so the outflow acceleration stalls (Figure 7(b)). Another source of this stalling is that the ions are pushed to finite x due to their double-welled χ's, and so their v_x couples with B_z (brown line in Figure 7(b)).

All in all, the initial outflow acceleration is due to the pickup-like mechanism. Later, both the Hall field E_y and the reconnection field E_z contribute to mean ion outflow acceleration. E_z does no net work on the ions in the z-direction.

ORCID iDs

Young Dae Yoon https://orcid.org/0000-0001-8394-2076
Paul M. Bellan https://orcid.org/0000-0002-0886-8782
Gunsu S. Yun https://orcid.org/0000-0002-1880-5865

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